

Beginner-Friendly C Array Practice Questions

This document is designed for **beginners learning arrays in C**. The questions progress from very simple concepts to slightly more logical operations. Students should attempt **dry runs on paper** before coding.

STEP 1: Understanding Arrays & Indexing

1. Read **N elements** into an array and print them.
2. Print array elements along with their **index values**.
3. Example: Index 0 -> 10
4. Print all array elements using a **for loop**.
5. Update the value at a **given index** and print the array.

Goal: Understand `arr[i]` and basic looping.

STEP 2: Reversing an Array

1. Reverse the array using **another array**.
2. Reverse the array **without using another array**.
3. Print the array from **last index to first index**.

Goal: Learn index movement (`i`, `n-1-i`).

STEP 3: Replacing Elements

1. Replace all **even numbers with 0**.
2. Replace a given number **X with Y** in the array.
3. Replace all **negative numbers with positive numbers**.

Goal: Use conditions and assignments inside loops.

STEP 4: Deleting Elements from an Array

1. Delete an element at a **given position**.
2. Delete an element by **value**.
3. Delete **all occurrences** of a given value.
4. Print the array after deletion.

Goal: Understand element shifting and array size reduction.

STEP 5: Inserting Elements into an Array

1. Insert an element at the **beginning** of the array.
2. Insert an element at the **end** of the array.
3. Insert an element at a **specific position**.

Goal: Learn backward shifting of elements.

STEP 6: Sorting Basics

1. Sort the array in **ascending order**.
2. Sort the array in **descending order**.
3. Print the array after **each pass** of sorting.

Goal: Understand nested loops and comparison logic.

STEP 7: Simple Logic Practice

1. Count how many times a number appears in the array.
2. Find the **largest and smallest** element.
3. Check whether the array is **already sorted**.
4. Copy one array into another.
5. Compare two arrays and check if they are equal.

Goal: Build confidence and logical thinking.

Teaching Notes

- Encourage students to **dry run** before coding.
- Ask questions like:
 - Which index is changing?
 - What happens to the array size?
 - Are elements shifting left or right?

This structured approach helps students build strong **array logic fundamentals in C**.